Jaypee University of Engineering and Technology, Guna
B.Tech. (Semester II)
Object Oriented Programming Lab (CS202)
Lab Exercise 10

1. You are developing a simple calculator program. Implement a method **calculate(int num1, int num2, char operator)** that performs basic arithmetic operations (addition, subtraction, multiplication, division). Handle exceptions for invalid operators and division by zero. Use custom exceptions where appropriate.

2. Write a C++ program that converts temperatures between Celsius and Fahrenheit. The program should first ask the user to choose whether they want to convert from Celsius to Fahrenheit or from Fahrenheit to Celsius. Then, it should take the temperature input from the user and validate that the input is not below absolute zero. Specifically, Celsius values should not be less than -273.15°C and Fahrenheit values should not be less than -459.67°F. You are required to create two separate functions—one for converting Celsius to Fahrenheit and another for converting Fahrenheit to Celsius.

   Use the formula,

   $F = (C * 9/5) + 32$ for converting Celsius to Fahrenheit and

   $C = (F - 32) * 5/9$ for converting Fahrenheit to Celsius.

   The program should call the appropriate function based on the user's choice and display the converted temperature. Make sure your program handles invalid inputs and provides appropriate error messages for temperatures below absolute zero.